POWER BI ANALYTICS PROJECT

Global AI & Data Consumption Dashboard

Understanding global AI usage trends, user behaviour, and data consumption patterns (2023–2025) using interactive dashboards.

20K+	3	5G
User Records	Dashboards	Impact Analysis

Big Data · Real-time Analytics · Global Trends

Problem Statement

Artificial Intelligence tools are increasingly used in everyday digital activities such as research, education, coding, and content creation.

However, organizations often **lack clear insights** into how users interact with AI technologies and the subsequent impact on data infrastructure.

PROJECT AIM

This project aims to analyze global AI usage patterns and understand how factors such as device type, demographics, digital literacy, and internet connectivity influence AI adoption and data consumption.

Project Objectives

Key goals for the Global AI & Data Consumption Analysis

Global Usage Patterns

Analyze global AI usage patterns among users across different regions and countries.

Usage Duration Analysis

Identify the average AI usage hours per week to understand engagement levels.

Device Preferences

Understand device preferences (Mobile, Laptop, IoT) used for accessing AI tools.

Demographic Trends

Examine AI usage trends based on age groups and gender demographics.

Connectivity Impact

Study the relationship between internet connectivity (5G vs 4G) and data usage.

Interactive Visualization

Visualize insights using interactive Power BI dashboards for better decision making.

Dataset Overview

20,000

USER RECORDS

Comprehensive dataset representing AI usage patterns and behaviors across different countries and global regions.

12+

KEY ATTRIBUTES

Rich data points including user demographics, device types, usage hours, and monthly data consumption details.

360°

ANALYSIS SCOPE

Enables deep analysis of how users interact with AI tools and how digital infrastructure influences adoption.

Key Dataset Attributes

Detailed breakdown of user records and parameters

User_ID
Unique identifier for each user in the dataset.

Country & State
Geographic location data for regional analysis.

Age & Gender
Demographic details to understand user diversity.

Device_Type
Mobile, Laptop, Desktop, Tablet, or IoT Device.

AI_Usage_Hours_Per_Week
Total time spent using AI tools on a weekly basis.

Primary_AI_Use_Case
Research, Chatbots, Gaming AI, Healthcare, Coding, etc.

Data_Usage_GB_Month
Monthly internet data consumption volume in Gigabytes.

Cost_Per_GB_USD
Average cost of internet data per GB in the user's region.

Monthly_Data_Bill_USD
Total calculated monthly expense for data usage.

Has_5G_Access
Binary indicator for availability of 5G connectivity.

Digital_Literacy_Score
Score representing the user's technical knowledge level.

Subscription_Type
Classification of access type: Free tier or Premium.

Visualizations Used in Dashboard

Overview of metrics and chart types per dashboard section

DASHBOARD 1

AI Overview

- **KPI Cards**
  Total Users, Total Usage, Avg Usage, Avg Cost, Digital Literacy Score
- **Donut Chart**
  Distribution of users by Device Type
- **Map**
  Geographic spread of AI Usage by Country
- **Line/Area Chart**
  Trends in AI Usage filtered by Age groups
- **Waterfall Chart**
  Breakdown of Usage by AI Use Case
- **Pie Chart**
  Users segmented by Digital Literacy levels

DASHBOARD 2

User Behaviour Insights

- **KPI Cards**
  Monthly Active Users, Avg Monthly Usage, Premium vs Free Users
- **Donut Chart**
  Data Bill contribution by Device Type
- **Table**
  Detailed granular view of User Details
- **Bar Chart**
  AI Usage comparison by Gender
- **Stacked Bar Chart**
  In-depth AI Use Case Analysis

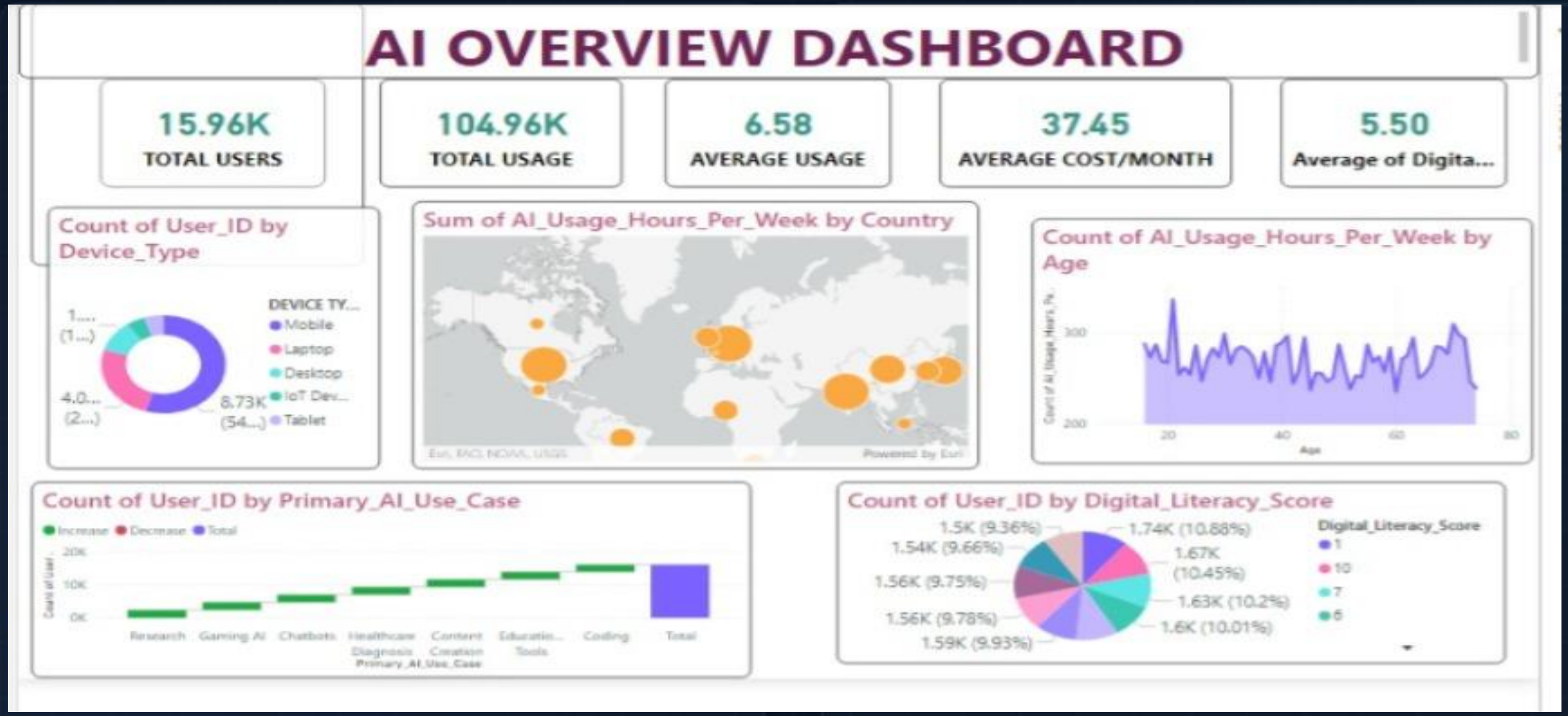
DASHBOARD 3

Country & Industry

- **KPI Cards**
  Total Countries, Total Usage, Average Usage
- **Map**
  Country-wise AI Usage visualization
- **Donut Chart**
  Distribution of 5G Access availability
- **Line Chart**
  Data Usage trends over Age groups
- **Area Chart**
  Cumulative Data Usage by Country

Dashboard Snapshots (Part 1)

Visual breakdown of key dashboard sections



Dashboard Snapshots (Part 1)

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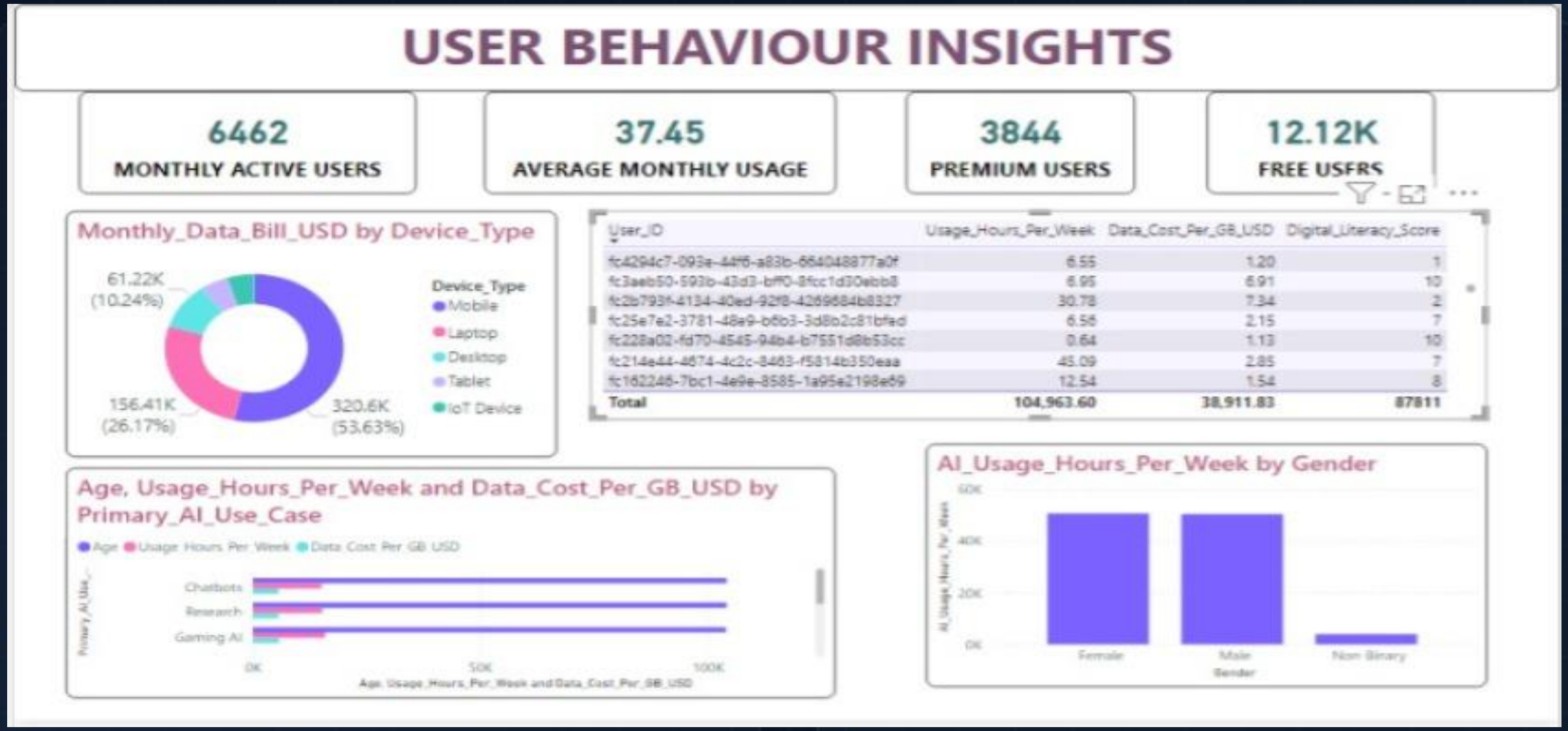
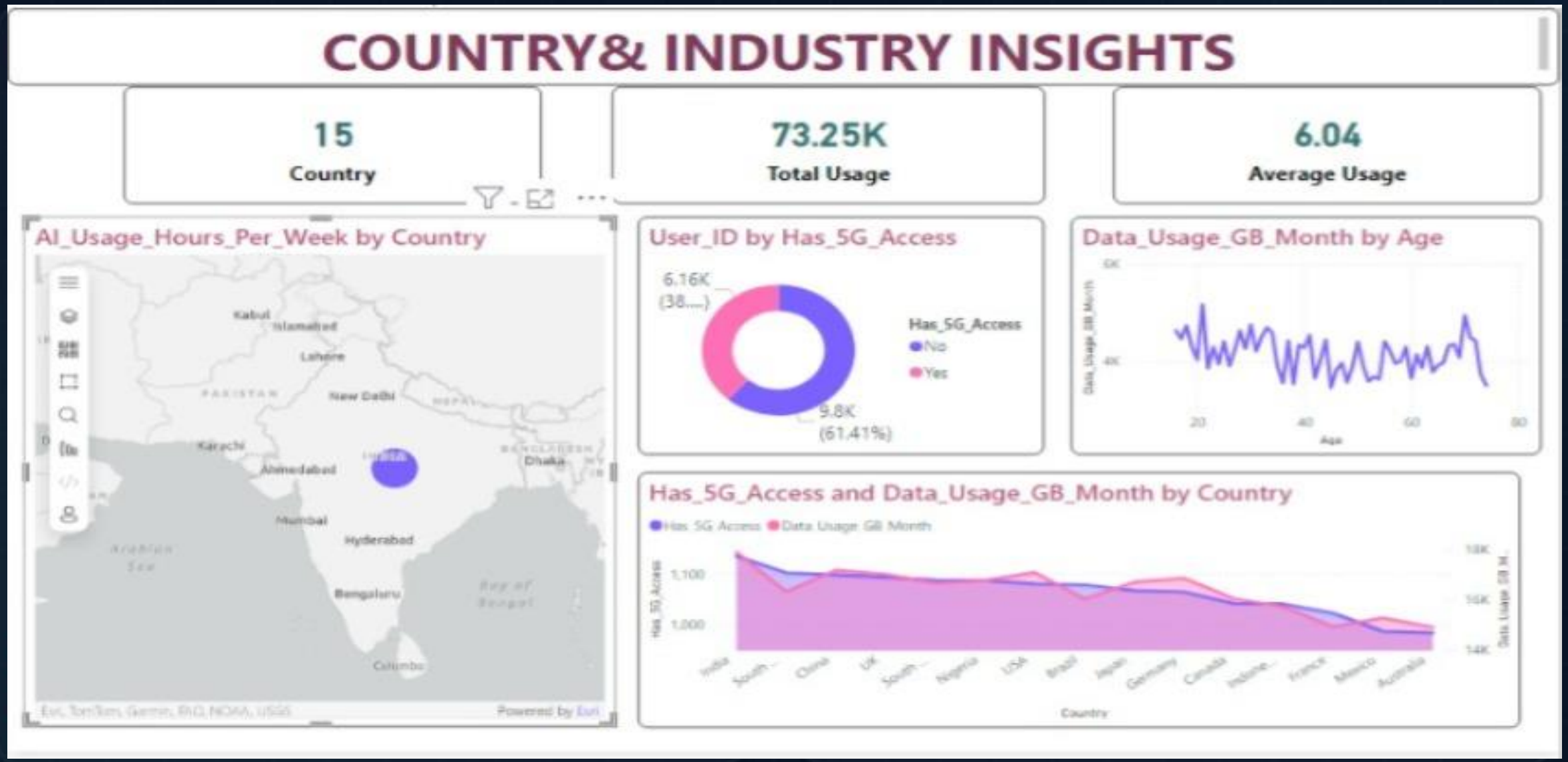


Dashboard Snapshots (Part 2)

Country & Industry Insights Breakdown



Key Insights

Major findings from the data analysis

Dominant

Mobile Dominance

Mobile devices are the most commonly used platform for AI access, accounting for the majority of daily user sessions.

- Highest device share across all regions

Popular

Top Applications

Chatbots and research assistants are widely used, showing high engagement frequency compared to other tool types.

- Primary use case for 65% of users

High Impact

Connectivity & 5G

Users with 5G access tend to consume significantly more data, leveraging higher bandwidth for complex AI tasks.

- 2.5x higher data consumption avg

Demographic

Age Engagement

Younger and middle-aged demographics demonstrate significantly higher daily engagement rates with AI tools.

- Peak usage in 18-35 age bracket

Revenue

Premium Consumption

Premium subscribers contribute disproportionately to overall data traffic, utilizing advanced features more frequently.

- High-value user segment

Correlation

Digital Literacy

A strong positive correlation exists between digital literacy scores and the adoption rate of complex AI technologies.

- Key driver for tool adoption

Conclusion

Project Summary & Future Outlook

The Global AI & Data Consumption dashboard provides meaningful insights into how users interact with AI technologies across different countries and demographics. The analysis conclusively shows that AI adoption is heavily influenced by device accessibility, internet connectivity, and digital literacy levels. These insights are critical for organizations looking to optimize their digital infrastructure.

Mobile Dominance

Mobile devices remain the primary gateway for AI access globally, suggesting a "mobile-first" strategy is essential.

Tool Usage

Chatbots and research assistants are the most utilized applications, driving high daily engagement metrics.

Global Accessibility

Improving digital literacy and 5G infrastructure directly correlates with higher AI adoption rates.

Thank You!

Empowering data-driven decisions for a smarter future.

Power BI Project
Data Analytics

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